CYBERMANAGEMENT

Modelo de datos

Diagrama entidad-relación y modelo relacional · Esquema inicial (migración e03f1d92d5f6)

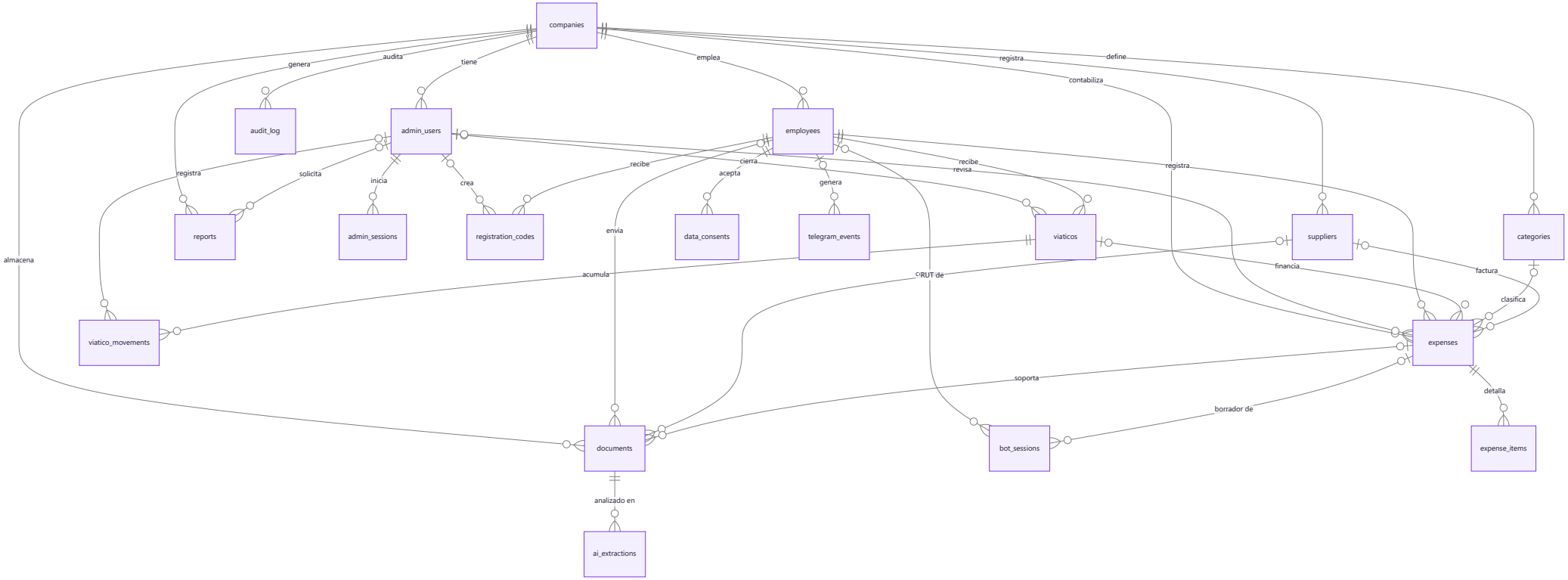
Refleja la migración `e03f1d92d5f6` (esquema inicial). Fuente de verdad: `backend/app/models/`.

Cómo leer las líneas (notación "pata de gallo"):

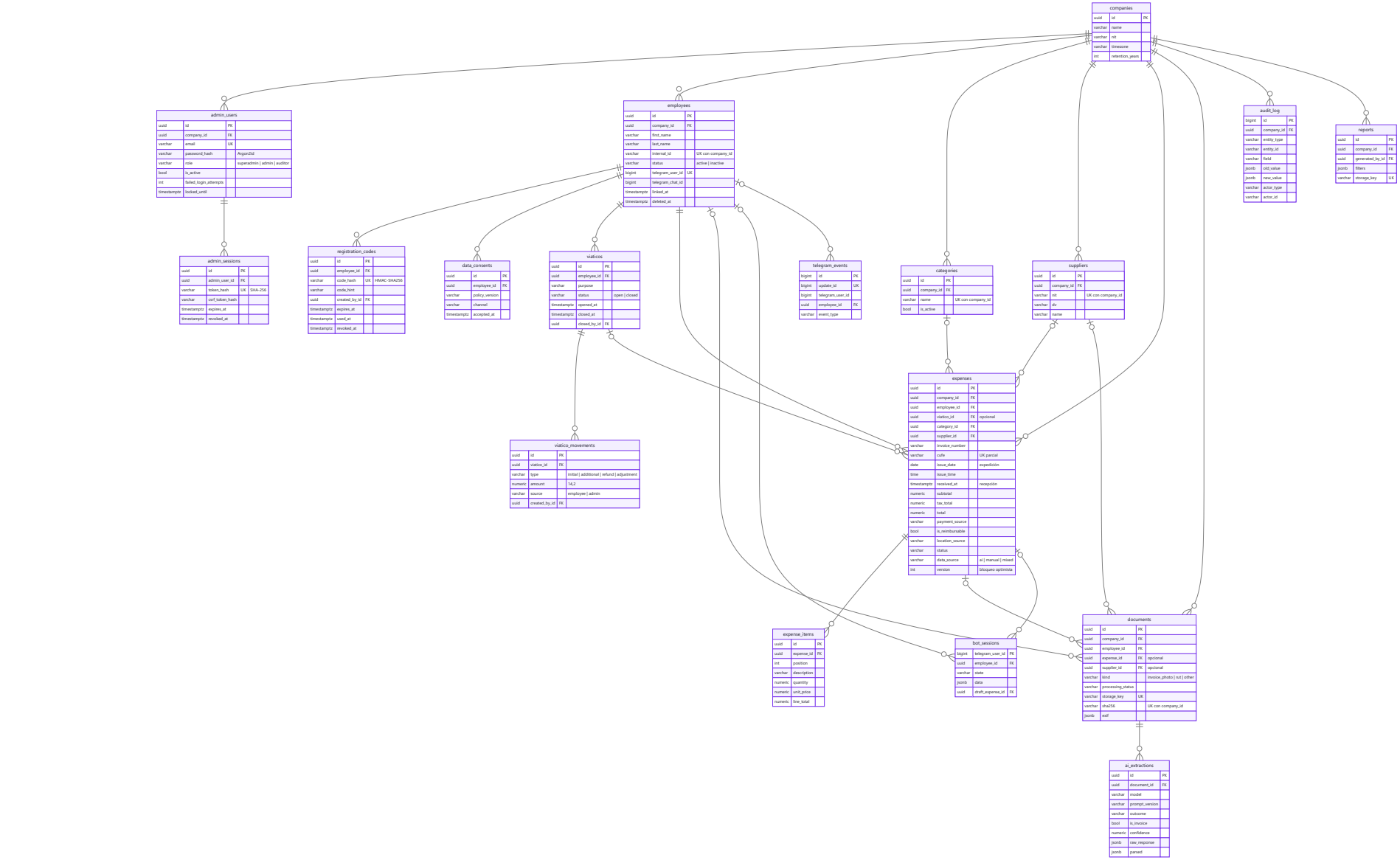
Símbolo	Significado
`||`	exactamente uno (obligatorio)
`|o`	cero o uno (opcional)
`o{`	cero o muchos

Ejemplo: `viaticos |o--o{ expenses` significa que un gasto pertenece a **cero o un** viático (puede pagarse con dinero propio), y un viático tiene **cero o muchos** gastos.

1. Vista general (entidades y relaciones)



2. Diagrama detallado



Para no saturar el diagrama detallado, las relaciones de auditoría hacia `admin_users` (`created_by_id`, `closed_by_id`, `reviewed_by_id`, `generated_by_id`) solo aparecen en la vista general.

3. Modelo relacional

Notación: **PK** clave primaria · **FK** → tabla referenciada · **UK** valor único · `?` admite nulos. Todas las tablas con `created_at` / `updated_at` los llenan automáticamente con la hora del servidor.

Empresa y administración

companies (**id** PK, name, nit?, timezone, retention_years, created_at, updated_at)

admin_users (**id** PK, company_id FK→companies, email UK, full_name, password_hash, role, is_active, last_login_at?, failed_login_attempts, locked_until?, created_at, updated_at)

admin_sessions (**id** PK, admin_user_id FK→admin_users *ON DELETE CASCADE*, token_hash UK, csrf_token_hash, created_at, expires_at, last_seen_at, ip_address?, user_agent?, revoked_at?)

Empleados

employees (**id** PK, company_id FK→companies, first_name, last_name, internal_id, status, telegram_user_id? UK, telegram_chat_id?, telegram_username?, linked_at?, deleted_at?, created_at, updated_at)

- UK (company_id, internal_id)

registration_codes (**id** PK, employee_id FK→employees *CASCADE*, code_hash UK, code_hint, created_by_id? FK→admin_users, created_at, expires_at, used_at?, used_by_telegram_user_id?, revoked_at?)

- UK parcial (employee_id) WHERE used_at IS NULL AND revoked_at IS NULL

data_consents (**id** PK, employee_id FK→employees *CASCADE*, policy_version, channel, accepted_at)

Viáticos

viaticos (**id** PK, employee_id FK→employees, purpose?, status, opened_at, closed_at?, closed_by_id? FK→admin_users, created_at, updated_at)

viatico_movements (**id** PK, viatico_id FK→viaticos *CASCADE*, type, amount, source, note?, created_by_id? FK→admin_users, created_at)

- CHECK amount <> 0 AND (amount > 0 OR type = 'adjustment')

Gastos

categories (**id** PK, company_id FK→companies, name, is_active)

- UK (company_id, name)

suppliers (**id** PK, company_id FK→companies, nit, dv?, name, created_at, updated_at)

- UK (company_id, nit)

expenses (**id** PK, company_id FK→companies, employee_id FK→employees, viatico_id? FK→viaticos, category_id? FK→categories, supplier_id? FK→suppliers, description?, invoice_number?, cufe?, issue_date?, issue_time?, subtotal?, tax_total?, total?, currency, received_at, payment_source?, is_reimbursable, location_lat?, location_lon?, location_text?, location_source, status, data_source, rejection_reason?, reviewed_by_id? FK→admin_users, reviewed_at?, deleted_at?, version, created_at, updated_at)

- CHECK total IS NULL OR total >= 0
- UK parcial (company_id, cufe) WHERE cufe IS NOT NULL AND deleted_at IS NULL
- UK parcial (company_id, supplier_id, invoice_number) WHERE ambos IS NOT NULL AND deleted_at IS NULL

expense_items (**id** PK, expense_id FK→expenses *CASCADE*, position, description, quantity, unit_price?, line_total?)

- CHECK quantity > 0

Archivos e IA

documents (**id** PK, company_id FK→companies, employee_id? FK→employees, expense_id? FK→expenses *ON DELETE SET NULL*, supplier_id? FK→suppliers, kind, processing_status, storage_key UK, mime_type, size_bytes, sha256, width?, height?, original_filename?, telegram_file_unique_id?, exif?, received_at, retention_until?, deleted_at?, created_at, updated_at)

- UK (company_id, sha256)

ai_extractions (**id** PK, document_id FK→documents *CASCADE*, model, prompt_version, outcome, is_invoice?, confidence?, raw_response?, parsed?, error?, latency_ms?, tokens_in?, tokens_out?, created_at)

Telegram

bot_sessions (**telegram_user_id** PK, employee_id? FK→employees *CASCADE*, state, data, draft_expense_id? FK→expenses *SET NULL*, updated_at, expires_at?)

telegram_events (**id** PK, update_id UK, telegram_user_id, employee_id? FK→employees *SET NULL*, event_type, received_at)

Trazabilidad

audit_log (**id** PK, company_id FK→companies, entity_type, entity_id, action, field?, old_value?, new_value?, actor_type, actor_id?, ip_address?, reason?, created_at)

- Índice (entity_type, entity_id). Sin FK hacia la entidad: es **polimórfica** (registra cambios de cualquier tabla) y debe conservarse aunque el registro original se archive.

reports (**id** PK, company_id FK→companies, generated_by_id? FK→admin_users, filters, storage_key UK, created_at)

4. Decisiones de diseño reflejadas en el modelo

Decisión	Por qué
`documents.expense_id` es opcional	Una foto se guarda **antes** de que exista el gasto, y una que no es factura nunca lo tendrá.
`expenses.viatico_id` es opcional	Si el empleado pagó con su dinero, no hay viático (`is_reimbursable = true`).
El saldo no es una columna	Se calcula con `viatico_movements` y `expenses` ; así no puede desincronizarse.
Fecha de expedición ≠ fecha de recepción	`issue_date` / `issue_time` vienen de la factura; `received_at` , de Telegram.
`ON DELETE CASCADE` solo en tablas "hijas" puras	Líneas de factura, sesiones o intentos de IA no tienen sentido sin su padre. Gastos, empleados y documentos usan borrado lógico.
`bot_sessions` usa `telegram_user_id` como PK	La conversación existe incluso antes de que el usuario se vincule como empleado.